DO NOW — RECALL

Lesson 109 · CPU Scheduling: FCFS, SJF and SRT

Name: _____ Class: _____ Date: _____

1. Name the routine executed when an interrupt is accepted.

2. An OR gate has inputs 0 and 1. What is the output?

3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

4. What is the time complexity of sequential search in the worst case?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Interrupt service routine (ISR).

2. Model answer

1.

3. Model answer

X is removed; Y remains at the front.

4. Model answer

O(n).

By the end, you can…

• First come, first served: non-preemptive queue in arrival order.

• Shortest job first: choose least burst time from ready jobs; non-preemptive.

• Shortest remaining time: preempt if a new job has a shorter remaining burst.

Vocabulary — know these terms

FCFS — Jobs run in first-arrival order.

SJF — Selects the shortest estimated ready job.

SRT — Preemptive shortest remaining time.

waiting time — Time spent ready but not executing.

NEW LEARNING

Read, annotate and explain

FCFS

First come, first served: non-preemptive queue in arrival order.

SJF

Shortest job first: choose least burst time from ready jobs; non-preemptive.

SRT

Shortest remaining time: preempt if a new job has a shorter remaining burst.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

FCFS: P1 arrives at 0 (burst 4), P2 at 1 (burst 2), P3 at 2 (burst 1). Find waiting times.

Model answer

Timeline P1 0–4, P2 4–6, P3 6–7. Waiting = start − arrival: P1 0, P2 3, P3 4. Average = 7/3 ≈ 2.33 units.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Distinguish non-preemptive and preemptive scheduling and give an example of each. [6]

Question 2

FCFS: A arrives 0/burst 3, B arrives 1/burst 2, C arrives 2/burst 4. Calculate each wait. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

SRT: A arrives 0/burst 5, B arrives 1/burst 2. Show the timeline and explain preemption. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

SRT never interrupts an executing process.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.